

Project Q3

Multi-Agent System (MAS) for Industrial IoT Data Refinement and Analytics

Background

AROL S.p.A. is a world leader in the design and production of advanced capping and filling machinery for bottles and jars, as well as capsule production. The company features high vertical integration: 95% of components and applications are developed in-house, ensuring total control over quality and production efficiency. With operational headquarters in Turin, other Italian cities, and key markets including the USA, Mexico, Brazil, China, and India, AROL provides a global presence serving various industrial sectors.

In recent years, AROL has undergone significant technological evolution, similar to that of the automotive sector. This shift has led to an increasing integration of advanced IT skills for centralized data management and machinery performance monitoring. AROL stands out for adopting advanced solutions across operating systems and system programming, web programming, modern user interfaces, machine learning, applied AI, and image recognition for quality control.

Project's Summary

The objective of this project is to design and implement a **Multi-Agent System (MAS)** capable of processing raw data streams from industrial equipment (specifically, **Equatorque**, a modern capping machine). Students will develop a decentralized architecture in which autonomous agents collaborate to ingest, clean, and analyze high-frequency telemetry data stored locally (previously synchronized from the Cloud).

Objectives

The main objectives are the following:

- **Data Ingestion & Local Synchronization:** Implement a mechanism to pull raw datasets from the Cloud and manage a local data persistence layer.
- **Agent-Based Data Cleaning:** Develop specialized "Cleaning Agents" that apply logic to filter noise, specifically addressing the mismatch between polling frequency and the actual production cycle (slower update rate).
- **Deduplication & Logic Filtering:** Create rules to identify and remove redundant entries while maintaining the integrity of the capping process timestamps.
- **Advanced Data Analytics:** Implement "Analytic Agents" that perform higher-level reasoning on the cleaned data (e.g., performance KPIs, anomaly detection, or trend analysis).
- **Scalability & Autonomy:** Demonstrate how the agent-based approach handles increasing data volumes better than traditional monolithic scripts.

Required Background

To successfully complete this project, students should possess, or be willing to acquire, knowledge in:

- **Distributed Systems:** Understanding of agent communication protocols and coordination models.
- **Industrial IoT (IIoT):** Familiarity with data structures generated by PLC-controlled machinery (capping torque, angle, success/failure flags).
- **Programming Skills:** Proficiency in Python or Java.
- **Data Management:** Basic SQL/NoSQL knowledge for local storage and data manipulation (Pandas/NumPy).

Working Environment

Students may work on a laptop or desktop, under Unix-like or Windows systems. The project can be naturally extended into an MSc thesis.

Deliverables

All projects must be delivered including:

- Source files.
- A README text file (plain ASCII) with instructions on how to compile and run the program, execution parameters, and dataset formats.
- A DOCUMENTATION file (Word, LaTeX, Markdown, etc.) containing:
 - main design choices (graph format, memory layout, CPU and GPU parallelization schemes, strategies for handling highly skewed degree distributions);
 - an experimental evaluation with tables and plots comparing:
- A set of OVERHEAD slides (PowerPoint or similar) for a 20-minute oral presentation of the projects.

Contact

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Additional Information

Additional details will be provided during the visit to Canelli on May 20th, 2026. It is recommended that at least one member of each group working on this activity attends the visit.